

KITALE, Kenya

WMO: 636610

Lat: 0.972N

Lon: 34.959E

Elev: 6070

StdP: 11.75

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	51.5	52.6	35.0	37.4	75.6	37.8	41.8	74.1	17.2	70.0	13.7	68.6	1.8	90	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.8	85.9	59.3	84.1	59.1	82.4	59.5	66.6	76.1	65.7	75.0	65.1	74.3	7.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.7	110.9	70.2	62.9	107.8	69.1	62.3	105.6	68.4	35.1	76.4	34.4	75.4	33.8	74.3	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	11.7	9.8	46.5	89.1	4.1	1.5	43.5	90.2	41.1	91.1	38.8	91.9	35.9	93.0		
(5)			43.4	71.4	2.8	3.4	41.4	73.8	39.8	75.8	38.2	77.7	36.1	80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.0	68.0	69.4	69.6	68.7	67.4	66.0	65.1	65.0	65.6	66.5	66.2	66.7
	DBStd	2.58	2.31	2.36	2.84	2.42	1.82	1.73	1.73	1.96	1.93	1.88	1.64	1.98
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	100	4	1	2	2	3	9	20	24	14	8	7	6
	CDD50	6202	558	543	606	562	539	480	467	464	469	510	487	517
	CDD65	828	97	124	144	114	77	39	22	22	33	53	44	59
	CDH74	6255	1088	1288	1263	666	327	151	97	107	285	279	229	475
	CDH80	1030	191	322	335	114	18	3	0	1	13	6	4	23
Wind	WSAvg	3.3	3.5	4.0	3.8	3.4	2.8	2.7	3.0	3.2	3.6	3.4	3.2	3.4
Precipitation	PrecAvg	48.7	0.9	2.2	3.5	6.6	6.6	3.9	5.0	5.8	3.5	3.7	3.3	1.4
	PrecMax	72.4	4.5	7.2	7.5	16.2	12.8	9.3	9.4	11.1	7.6	8.9	14.4	5.2
	PrecMin	34.7	0.0	0.0	0.2	1.8	1.5	0.5	0.4	0.1	0.1	0.6	0.2	0.1
	PrecStd	7.8	1.2	2.2	2.1	2.8	2.7	2.0	2.5	2.5	2.1	2.2	3.0	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.1	87.9	88.1	86.3	82.2	79.8	78.4	78.9	82.0	80.7	80.4	82.3
		MCWB	58.8	58.5	58.9	60.9	64.6	64.3	63.8	64.2	61.2	62.2	63.2	60.3
	2%	DB	83.9	85.5	85.9	83.1	79.3	77.3	76.5	76.8	79.2	78.8	78.3	80.3
		MCWB	58.8	58.7	59.3	61.0	64.3	63.9	63.6	63.4	61.9	62.4	62.4	60.3
	5%	DB	82.2	83.8	83.9	80.5	77.4	75.5	75.0	75.0	77.3	77.1	76.7	78.6
		MCWB	58.9	58.4	59.2	61.8	64.1	63.4	63.2	63.1	62.3	62.5	62.3	60.2
	10%	DB	80.2	82.0	81.2	78.1	75.6	73.8	73.2	73.2	75.2	75.4	75.0	76.9
		MCWB	59.0	58.4	60.0	62.4	63.9	63.2	62.8	62.8	62.4	62.4	62.1	60.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.9	64.9	65.7	67.3	67.5	66.9	66.3	66.2	66.4	66.3	66.5	65.9
		MCDB	76.1	76.0	75.8	77.1	77.5	76.1	75.5	76.2	76.8	75.6	75.4	74.4
	2%	WB	63.2	63.4	64.3	65.8	66.3	65.6	64.8	64.8	64.9	65.0	65.0	64.5
		MCDB	74.4	74.6	74.8	74.7	75.9	74.6	73.9	74.1	74.8	74.4	73.7	73.4
	5%	WB	62.1	62.3	63.4	64.8	65.4	64.6	63.9	63.9	63.7	64.0	64.1	63.3
		MCDB	73.3	73.4	73.6	73.5	74.4	73.3	72.6	72.7	73.1	73.2	72.7	72.1
	10%	WB	61.1	61.4	62.6	64.0	64.6	63.6	63.0	62.9	62.8	63.1	63.2	62.3
		MCDB	72.7	72.8	72.5	72.2	73.0	71.7	71.3	71.2	71.7	71.9	71.3	71.0
Mean Daily Temperature Range	5% DB	MDBR	26.9	26.8	24.1	20.4	18.8	19.0	18.9	19.2	21.2	20.1	19.8	22.6
		MCDBR	29.6	29.9	28.7	24.5	21.9	21.0	21.2	21.7	24.7	22.6	22.5	25.7
		MCWBR	10.1	9.3	8.1	7.3	8.8	9.3	9.6	9.8	10.3	9.0	9.2	9.8
	5% WB	MCDBR	25.1	25.1	22.0	19.5	19.5	19.6	19.8	20.3	21.5	20.1	19.6	21.0
		MCWBR	9.6	8.9	7.7	7.6	8.7	9.4	9.7	9.7	10.0	9.0	9.1	9.2
Clear-Sky Solar Irradiance	taub	0.334	0.354	0.362	0.352	0.363	0.418	0.431	0.405	0.369	0.357	0.343	0.337	
	taud	2.528	2.465	2.463	2.542	2.503	2.303	2.251	2.333	2.435	2.476	2.532	2.531	
	Ebn at Noon	315	310	304	301	290	270	268	281	298	305	310	313	
	Edh at Noon	34	37	37	34	34	40	43	41	38	36	34	34	
All-Sky Solar Radiation	RadAvg	1990	2077	2029	1875	1839	1717	1656	1748	1948	1885	1806	1884	
	RadStd	137	114	134	116	64	93	83	87	76	69	100	119	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page